

Assignment 01

Handed out: 09/05/2026

Due by 1 PM EST on Saturday, 09/12/2026

The purpose of this assignment is to demonstrate how easy it is to run TensorFlow (Keras) code. You are also asked to create and manipulate Conda virtual environments and perform the basic operations in a Jupyter notebook. All those skills might be useful in some situations. You are also asked to demonstrate use of a notebook in Google Colab.

It is our advice that you work with Anaconda and use `conda` to create one virtual environment with TensorFlow (Keras). If you are short on space you can work with MiniConda. If you know what you are doing, choose your environment management tool as you please.

The virtual environment for this course should have Python 3.8 or higher and TensorFlow 2.10 or higher. The latest version of TensorFlow is 2.21. If you are confident in what you are doing, use any version of Python and any version of TF equal or higher than 2.10. On Windows machines with a GPU card, we recommend working with TF 2.10. The code in our examples works with that version of TensorFlow and later. When installing TensorFlow 2.10, please create virtual environment with Python 3.8, not a higher version. Add `numpy 1.23.5` with command `conda install numpy==1.23.5`. Then, add Jupyter with command `conda install jupyter`.

Virtual environments are needed when you work on your local computer. For higher computational needs you will switch to Google Colab or one of GPU Cloud Providers.

Problem 1. Install Anaconda or MiniConda. Open Anaconda prompt. Use `Conda` to create new virtual environment. When creating the environment, specify that you want Python 3.8 or higher. Capture the dialog when creating and activating the environment. Once inside the new virtual environment, open the Python prompt and examine the version of Python. Open a regular command prompt not within just created virtual environment. Examine system's Python if any. Test whether 'pip' is present in the environment. If `pip` is not present, install the latest version. Find and run `conda` command that lists existing virtual environments. List existing virtual environments. Use the `pip` command to list packages installed in your virtual environment (15%)

Problem 2. In your virtual environment install the latest version of TensorFlow if working on Linux, MacOS, or WSL. Use TensorFlow 2.10 if working on Windows. Open a Python prompt and examine the version of TensorFlow installed. Again, list Python packages installed in your environment. (15%)

Problem 3. Inside your virtual environment install Jupyter (notebook), matplotlib, pandas, and numpy packages. Open Jupyter notebook:

```
$ jupyter notebook
```

Name your notebook `e89_LastName_FirstName_HW01`. Create a markup cell and copy all the results collected in Problem 1. Create another markup cell and copy all the results from Problem 2. The objective is to submit one notebook as the solution to this problem set. Demonstrate that you could import tensorflow as tf. Create two simple tensors of dimensions 2x2 and add them up. Please add two tensors using mathematical shorthand symbol “+” and then operator `tf.add()`. Print the values of the resulting tensors. (15%)

Problem 4. Copy the code from the attached Jupyter notebook `cpu_vs_gpu_test.ipynb` into your solution notebook. Copy and run each cell separately. Copy markup cells as well. Run all cells. It is alright if your machine does not have CUDA support. Just report your results. (15%)

Problem 5. Run tests from the previous problem #4 in Google Colab. https://colab.research.google.com/?utm_source=scs-index First measure the execution speed with CPU. Then create a paid subscription (Colab Pro \$10 per month, <https://colab.research.google.com/signup>) and use GPUs. Copy results of both tests in your local notebook: `problemset01`. Again, the objective is to have only one notebook for submission. (20%)

Problem 6. Run every cell in the attached notebook `Lec01_Introduction_Simple_Example.ipynb` both locally and in Google Colab.. Choose 4 random samples from the test set. Report what digits your model recognizes for those samples. Repeat the experiment with a reduced training set of 10,000 samples. Report on differences with the original run in any. (20%)

Formatting Requirements:

Solution of this assignment should preferably be submitted as a single Jupyter notebook (ipynb file), as well as an HTML image of that notebook. To generate the HTML image, use the following `jupyter` command:

```
$ jupyter nbconvert e89_LastName_FirstName_HW01.ipynb --to html
```

Please provide meaningful description of actions in all cells of your notebook. Include those descriptions either as Python comments (#) or contents of the adjacent markdown cells.

Please name your notebook as `e89_LastName_FirstName_HW01.ipynb`. We need your name in the file name. Please add your name in a markup cell in the notebook itself. Upload to Assignment 01 in Canvas.